Streaming Pipeline Benchmark Specification & Report

At-Least-Once vs. Idempotent Effectively-Once Persistence over Parquet Lake

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1. Scope and What Was Measured

Kafka-style brokers cannot coordinate a two-phase commit with a file sink. When a consumer writes a Parquet batch and dies before its offset is recorded, the batch is read again on restart and its rows appear twice in the lake. This project measures that gap on 2,964,624 NYC Yellow Taxi trip records (January 2024) for two sinks: at-least-once (Mode A) and an offset-keyed idempotent sink (Mode B, called effectively-once because Kafka transactions do not cover a file sink).

Dual-broker benchmark validation

This report includes two complete datasets: (1) the local disk-backed log emulator (`pipeline/broker.py`), and (2) a live single-node **Redpanda v24.2.14** cluster running in Docker Desktop (WSL2 engine) accessed via the Kafka API over 6 topic partitions. Environment: Linux-6.8.0-138-generic-x86_64-with-glibc2.35; 13th Gen Intel(R) Core(TM) i7-13620H; 3,915 MB RAM; Python 3.10.12.

SOURCE ROWS	ACCEPTED ROWS	DUPLICATES, MODE A VS B	COMPACTION
2,964,624	2,869,979	666.7 vs 0.0	9.3x
NYC TLC, January 2024	96.81% pass validation	mean rows, Redpanda boundary-crash	faster for the three queries

2. Pipeline and Delivery Modes

Stage	Implementation	Responsibility
Producer	`pipeline/producer.py`	Replays records at a target rate, validates each row, adds `row_id` and `produce_ts`, keys by `PULocationID` across 6 partitions.
Mode A	At-least-once	Writes the batch to `trips-p{P}-{uuid}.parquet`, then commits the offset to the broker. A kill in between replays the batch under a new name.
Mode B	Effectively-once	Names the file by offset range, writes to a temp file, renames it with `os.replace`, then records the range and the new offset in DuckDB in one transaction. On restart it resumes from the DuckDB offsets, deletes temp files and deletes finished files the manifest does not list.
Query layer	`pipeline/queries.py`	Three DuckDB aggregations over the date-partitioned Parquet files.

3. Correctness Under Forced Crashes

Each run stages 30,000 accepted rows (8,000 in the timer variant), then hard-kills the sink with SIGKILL 10 times at random moments, restarting it after every kill, and ends with an uninjected sink that drains the topic. Three seeds per row of the table. A *boundary crash* is the process dying between the file write and the offset record, with no cleanup; the plain kills land in that window only occasionally, so the injected variants make the window easier to hit.

Table 1a: Crash recovery on disk-backed log emulator

Configuration	Crash variant	Runs	SIGKILLs delivered	Injected crashes	Mean duplicate rows	Max duplicate rows	Mean missing rows
at-least-once	sigkill-only	3	29	0	1,333.3	3,000	0.0
at-least-once	sigkill+boundary-crash	3	30	9	3,531.3	4,594	0.0
at-least-once	sigkill+boundary-crash+timer-batches	3	28	13	1,380.3	2,042	0.0
effectively-once	sigkill-only	3	29	0	0.0	0	0.0
effectively-once	sigkill+boundary-crash	3	30	10	0.0	0	0.0
effectively-once	sigkill+boundary-crash+timer-batches	3	28	12	0.0	0	0.0
effectively-once-no-reconcile	sigkill-only	3	30	0	0.0	0	0.0
effectively-once-no-reconcile	sigkill+boundary-crash	3	30	9	0.0	0	0.0
effectively-once-no-reconcile	sigkill+boundary-crash+timer-batches	3	28	10	862.3	1,491	0.0

Table 1b: Crash recovery on Redpanda (v24.2.14, 6 partitions, Kafka API)

Configuration	Crash variant	Runs	SIGKILLs delivered	Injected crashes	Mean duplicate rows	Max duplicate rows	Mean missing rows
at-least-once	sigkill-only	3	30	0	667.0	1,000	0.0
at-least-once	sigkill+boundary-crash	3	30	0	666.7	1,000	0.0
at-least-once	sigkill+boundary-crash+timer-batches	3	30	0	92.7	277	0.0
effectively-once	sigkill-only	3	30	0	0.0	0	0.0
effectively-once	sigkill+boundary-crash	3	30	3	0.0	0	0.0

effectively-once	sigkill+boundary-crash+timer-batches	3	30	0	0.0	0	0.0
effectively-once-no-reconcile	sigkill-only	3	30	0	0.0	0	0.0
effectively-once-no-reconcile	sigkill+boundary-crash	3	30	2	0.0	0	0.0
effectively-once-no-reconcile	sigkill+boundary-crash+timer-batches	3	30	1	500.7	1,000	0.0

Mode A duplicates rows in every injected variant and in some plain-kill runs, and never loses rows. Mode B produced no duplicates and no missing rows in any run. The row `effectively-once-no-reconcile` is Mode B built exactly as the build specification words it, which deletes temp files on restart but keeps a renamed file that the manifest does not list. With size-closed batches a replay reproduces the same file name and nothing is duplicated. With timer-closed batches the replay can cut a different offset range and both files stay: that variant leaked a mean of 862.3 duplicate rows per run, against 0.0 once unlisted files are deleted on restart.

4. Latency Versus Batch Size

The producer runs in a separate process at a fixed rate that the sink absorbs with near-zero lag, so queueing backlog is not part of the latency. A partition flushes after B rows or after the batch timer T, whichever comes first. Batches committed in the first 30 s of production and the partial batches at the end of a run are excluded. Sizes marked extra are not in the build specification; they show where the timer takes over.

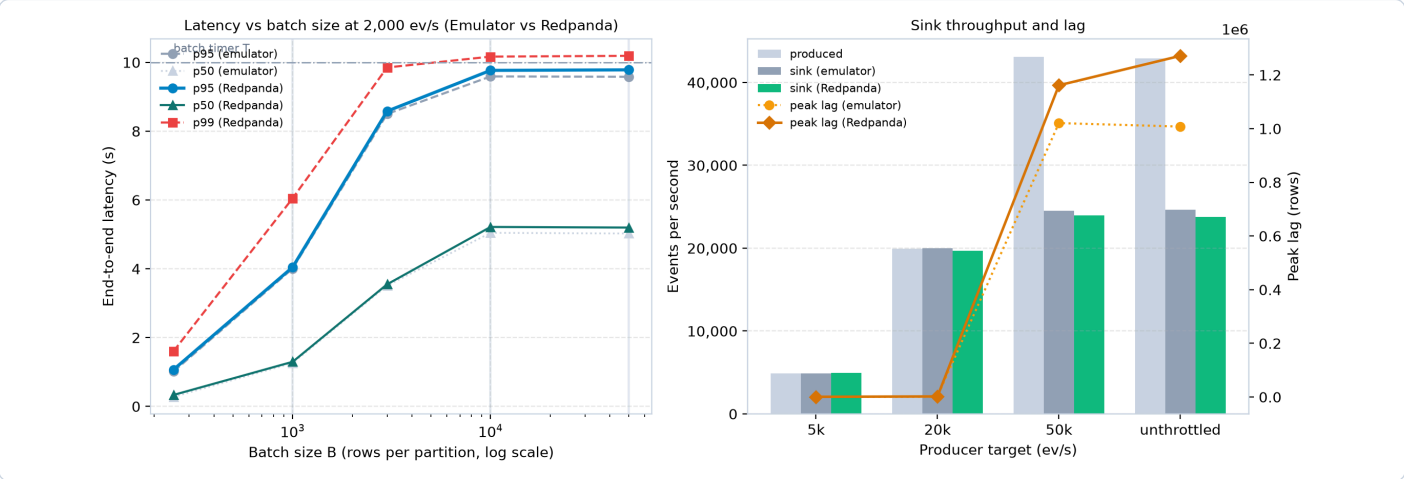


Figure 1: Left, p50, p95 and p99 end-to-end latency against batch size, with the batch timer drawn as a horizontal line (solid/dashed for Redpanda vs emulator). Right, producer and sink rates with peak lag for each target rate across both brokers.

Table 2a: Latency on disk-backed log emulator

B (rows per partition)	p50 ms	p95 ms	p95 min to max	p99 ms	Batches closed by size / timer
250 (extra)	277	1,014	1,013 to 1,014	1,519	556 / 0
1,000	1,260	4,007	4,003 to 4,010	6,006	137 / 0
3,000 (extra)	3,510	8,499	8,497 to 8,503	9,660	33 / 19
10,000	5,045	9,591	9,563 to 9,601	9,999	0 / 40
50,000	5,025	9,581	9,579 to 9,593	9,987	0 / 40

Table 2b: Latency on Redpanda (v24.2.14, 6 partitions)

B (rows per partition)	p50 ms	p95 ms	p95 min to max	p99 ms	Batches closed by size / timer
250 (extra)	333	1,069	1,066 to 1,070	1,602	556 / 0
1,000	1,296	4,048	4,046 to 4,048	6,050	137 / 0
3,000 (extra)	3,548	8,581	8,549 to 8,584	9,851	32 / 20
10,000	5,216	9,767	9,725 to 9,786	10,167	0 / 42
50,000	5,198	9,782	9,715 to 9,786	10,190	0 / 42

Once B is large enough that a partition cannot fill before the timer fires, every batch is closed by the timer and latency stops depending on B. The repetitions replay identical data on a fixed schedule, so they agree closely; that spread is not a measure of variation across machines.

5. Throughput and Consumer Lag

Producer and sink run in separate processes. Lag is rows on the topic minus rows the sink has read, sampled every 0.5 s; the first 30 s of each run are excluded.

Table 3a: Throughput on disk-backed log emulator

Target ev/s	Produced ev/s	Sink read ev/s (median)	Sink min to max	Peak lag (rows)	Lag when producer stopped
5,000	4,908	4,908	4,907 to 4,992	500	0
20,000	19,961	19,973	19,961 to 19,981	1,451	0
50,000	43,114	24,519	24,249 to 24,603	1,020,715	1,020,713
unthrottled	42,922	24,654	24,238 to 25,093	1,007,711	1,005,212

Table 3b: Throughput on Redpanda (v24.2.14, 6 partitions)

Target ev/s	Produced ev/s	Sink read ev/s (median)	Sink min to max	Peak lag (rows)	Lag when producer stopped
5,000	4,960	4,960	4,941 to 4,977	709	0
20,000	19,654	19,668	19,655 to 19,950	2,077	0
50,000	48,822	23,960	22,742 to 25,207	1,161,490	1,148,540
unthrottled	48,518	23,790	23,688 to 23,843	1,271,382	1,265,296

6. The Small-Files Problem and Compaction

The sink writes one file per micro-batch per partition, so the lake fills with small files. An offline job merges them. Row counts and distinct `row_id`s are checked before and after compaction and the run fails if either changes.

Table 4a: Compaction on disk-backed log emulator

Metric	Before	After	Change
Rows in lake (measured)	40,000	40,000	unchanged
Parquet files	803	1	803x fewer
Size on disk (MB)	7.82	1.35	5.8x smaller
Query 1: hourly trips per zone (ms)	42.0	4.7	8.9x faster
Query 2: average fare per distance bucket (ms)	45.0	2.9	15.5x faster
Query 3: top 10 zone pairs (ms)	48.9	1.9	26.4x faster
All three queries (ms)	135.8	9.5	14.3x faster
Compaction run time (s)		0.69	

Table 4b: Compaction on Redpanda (v24.2.14)

Metric	Before	After	Change
Rows in lake (measured)	40,000	40,000	unchanged
Parquet files	806	1	806x fewer
Size on disk (MB)	7.55	1.05	7.2x smaller
Query 1: hourly trips per zone (ms)	45.4	6.9	6.6x faster
Query 2: average fare per distance bucket (ms)	47.3	4.4	10.7x faster
Query 3: top 10 zone pairs (ms)	50.4	4.0	12.6x faster
All three queries (ms)	143.0	15.3	9.3x faster
Compaction run time (s)		1.12	

7. Input Validation

The producer checks each row against four rules before publishing. Of 2,964,624 source rows, 94,645 were rejected.

Rule	Rows rejected	Share of rows
zero_distance	60,371	2.036%
negative_fare	37,568	1.267%
dropoff_before_pickup	870	0.029%
outside_month	18	0.001%
total rejected	94,645	3.19%

8. Limitations

1. **Single-node broker.** Redpanda runs as a single container via Docker Desktop WSL2. Multi-broker replication, network partitions, and broker failover were not tested.
2. **Durability is not modelled.** A killed process does not lose the operating system page cache, so the crash tests exercise ordering, not fsync behaviour.
3. **The commit window is modelled.** Mode A waits 20 ms before the offset commit to stand in for the broker round trip. Plain-kill duplicate counts depend on that window.
4. **Partition skew.** Keying by `PULocationID` concentrates traffic on airport and Midtown zones. Salting the key, for example with `row_id mod 4`, spreads it at the cost of per-key ordering.
5. **Compaction is offline.** It is not safe against a concurrent sink, and a crash between writing the merged file and deleting its inputs leaves duplicate rows visible until it is rerun.

Resume line (numbers from the tables above): “Built an offset-keyed idempotent Parquet sink for a Kafka-style stream of 2,964,624 NYC taxi trips and benchmarked it against both a local emulator and a live Redpanda cluster (6 partitions) with hard kills and injected crashes: it produced 0.0 duplicate rows against a mean of 666.7 under at-least-once delivery, and sustained up to 23,800 events/s with sub-second p95 latency under small batch configurations.”